电磁感应与电磁场

一、电磁感应定律:

$$\mathcal{E} = -\frac{d\Phi}{dt} \quad *$$

二、楞次定律:

闭合回路感应电流的方向,总是使得它所激发的磁场来阻止引起感应电流的变化

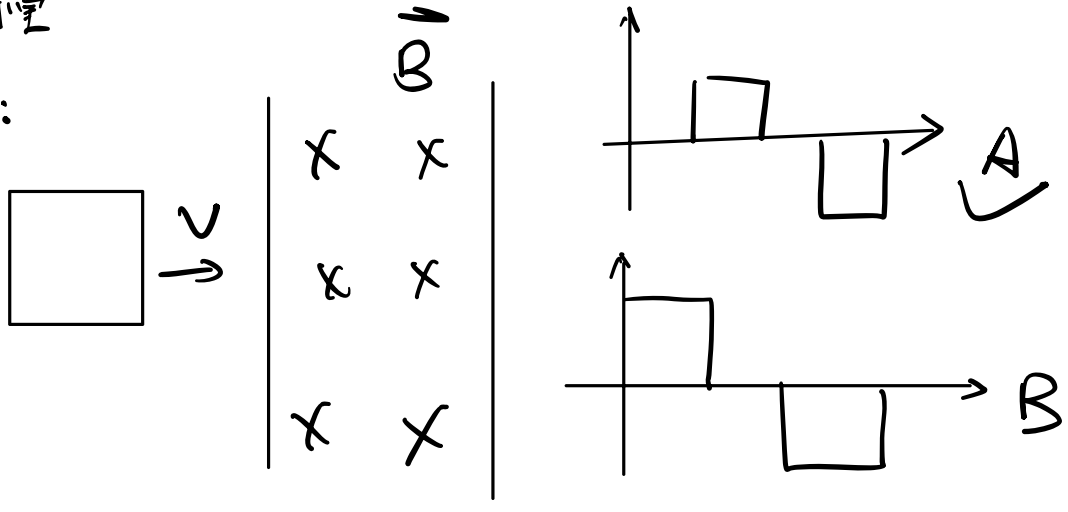
三、电动势

$$\mathcal{E} = \oint_L \vec{E_k} \cdot d\vec{l}$$

$\mathcal{E}$: "标量"

$\vec{E_k}$: 非静电力

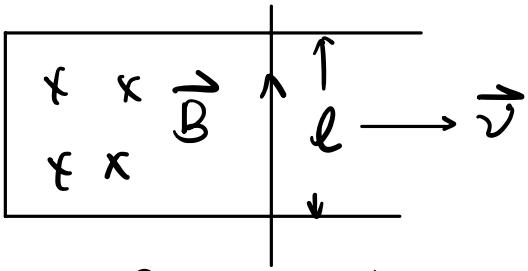
例1:



四、动生电动势/感生电动势

动: $\mathcal{E} = BLv$

$\mathcal{E} = \vec{v} \times \vec{B} \cdot L$

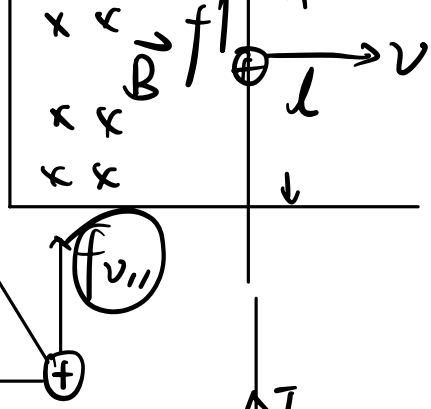


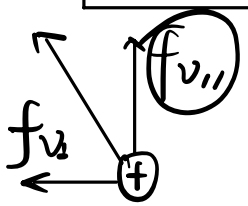
$$\mathcal{E} = \int_-^+ \vec{E_k} \cdot d\vec{l} = \int_a^b (\vec{v} \times \vec{B}) \cdot d\vec{l}$$

微观解释:

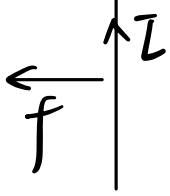


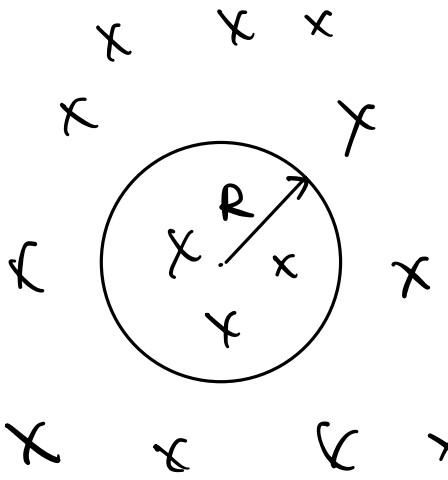
$f_{洛}: q\vec{v}\times\vec{B}$





感:





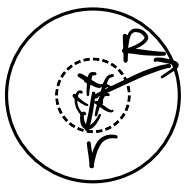
$\frac{d\vec{B}}{dt}=k$

$\mathcal{E}=-\frac{d\Phi}{dt}$

$\Phi=B\cdot\pi R^2$

$\Phi(t)=B(t)\pi R^2$

$\mathcal{E}=-\frac{d\Phi}{dt}=-k\pi R^2$

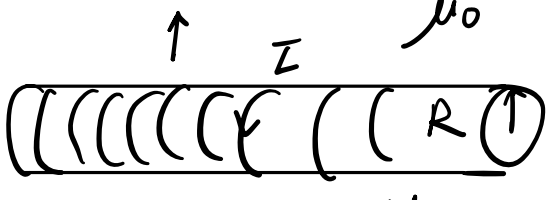


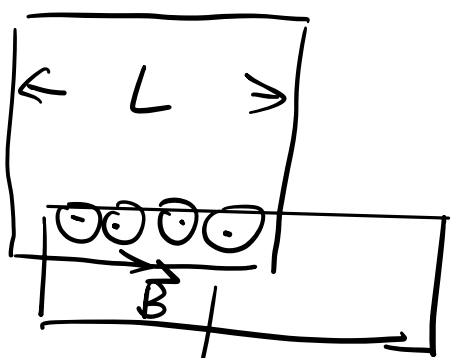
$\mathcal{E}=-\frac{d\Phi}{dt}=-k\pi r^2$

五、自感/互感

自: I $B\propto I$

自感系数 $\Phi=LI$





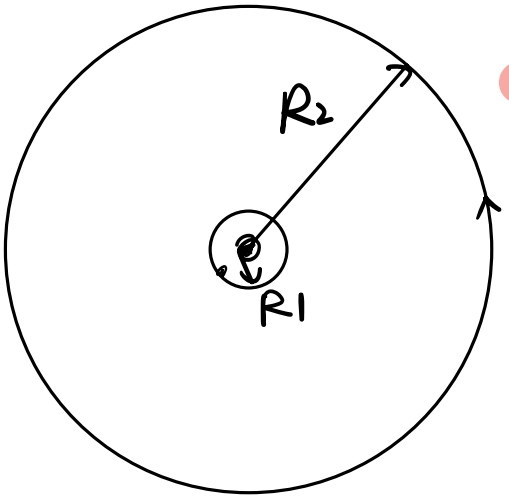
$\oint\vec{B}\cdot d\vec{l}=\mu_0\Sigma I$

$B\cdot L=\mu_0\cdot nLI$

$\Phi = B\pi R^2 = \mu_0 n I \cdot \pi R^2 = LI$ $\quad B = \mu_0 n I$

互:

$L = \mu_0 n \pi R^2 \qquad \Phi = MI$



$R_2 \gg R_1$ $\quad M = ?$ $\quad \Phi_{21} = ?$

$\Phi_{21} = \frac{\mu_0 I \pi}{2R_2} \cdot \pi R_1^2$

$= MI$

$dB = \frac{\mu_0}{4\pi} \frac{I \cdot dl}{r^2}$

$M = \frac{\mu_0 \pi^2 R_1^2}{2R_2}$

$B = \frac{\mu_0}{4\pi} \cdot \frac{I \cdot 2\pi R_2}{R_2^2}$

$= \frac{\mu_0 I \cdot \pi}{2R_2}$

相关

(1) 自感磁能

$i \quad \Delta q$

$dW = \varepsilon_L dq = -L \cdot \frac{di}{dt} \cdot dq$

$= -L di \cdot \frac{dq}{dt} = -L di \cdot i$

$dW = -Li\,di$ $\quad i\ (0 \to I)$

$W = \int_0^I Li\,di$

$= \frac{1}{2} LI^2$

$F \Delta t = m \Delta v$

$F = m \frac{\Delta v}{\Delta t} = ma$

$Fdx = dW$

$L \longleftrightarrow M$

$I \longleftrightarrow v$

$q \longleftrightarrow x$

$\frac{1}{C} \longleftrightarrow k$

$U \longleftrightarrow F$

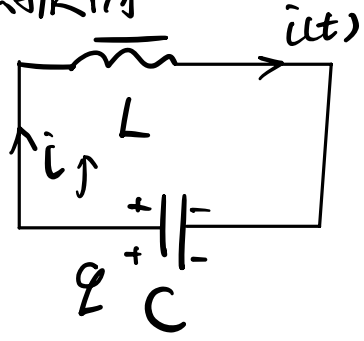
$$dW = m\frac{\Delta v}{\Delta t}\cdot \Delta x = m\Delta v \frac{\Delta x}{\Delta t}$$

$$= m\Delta v \cdot v$$

$$W = \int_0^v mv\,dv = \frac{1}{2}mv^2$$

(2)

电磁振荡



$$-L\frac{di}{dt} = \frac{q}{C}$$

$$\frac{q}{C} + L\frac{di}{dt} = 0$$

$$L\frac{di}{dt} + \frac{q}{C} = 0$$

$$\frac{di}{dt} + \frac{1}{CL}\cdot q = 0$$

$$\ddot{q} + \frac{1}{CL}q = 0$$

$$\omega = \sqrt{\frac{1}{CL}} \qquad T = \frac{2\pi}{\omega} = 2\pi\sqrt{LC}$$

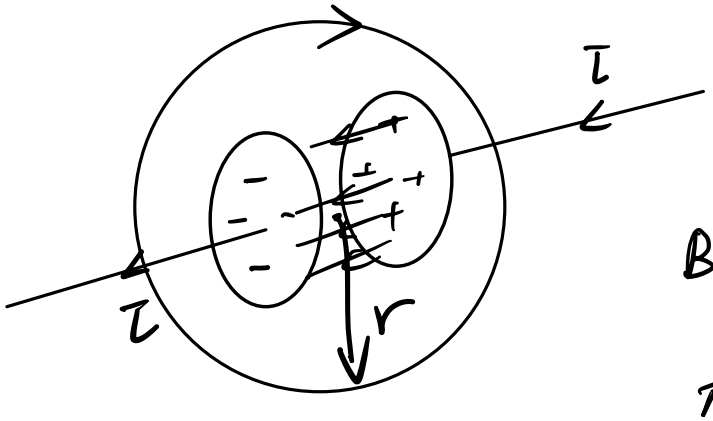
ω^2

$T = 2\pi\sqrt{\frac{m}{k}}$

$$\ddot{x} + \frac{k}{m}x = 0$$

$$\ddot{x} + \omega^2 x = 0$$

六、位移电流

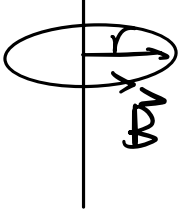


$$B\cdot 2\pi r = \mu_0 \Sigma I = 0$$

存在 → 电场

$$\oint \vec{B}\cdot d\vec{l} = \mu_0 \Sigma I$$

$$I_d = \frac{d\Phi_e}{dt} = \iint_S \frac{\partial \vec{D}}{\partial t}\cdot d\vec{S}$$



$B \cdot 2\pi r = \mu_0 I$

$B = \frac{\mu_0 I}{2\pi r}$

修正后的安培环路定理:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 (I_c + I_d) = \mu_0 I_c + \mu_0 \iint_S \frac{\partial \vec{D}}{\partial t} \cdot d\vec{S}$$

麦克斯韦方程组: (积分形式)

① $\iint \vec{D} \cdot d\vec{S} = q_0$ $(\vec{D} = \varepsilon_r \varepsilon_0 \vec{E})$

② $\oint_L \vec{E} \cdot d\vec{l} = -\iint \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S} = -\frac{d\Phi}{dt}$

③ $\iint \vec{B} \cdot d\vec{S} = 0$ $(\vec{B} = \mu_r \cdot \mu_0 \vec{H})$

④ $\oint_L \vec{B} d\vec{l} = \mu_0 \iint_S (\vec{j}_c + \frac{\partial \vec{D}}{\partial t}) d\vec{S} = \mu_0 I_c + \mu_0 I_D$ $(j_c = \frac{I}{S})$